



Examining Autonomous Robotic Navigation Hardware:

AI-Driven and Manually Controlled Systems

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Introduction

Autonomous robots are complex systems that perform a wide range of functions from perception and decision-making to execution. Therefore, they are deployed for various applications and operate in highly uncertain environments, such as streets and factories. TinyML brings this autonomy to the edge, allowing for real-time model inference and decision-making directly on the device despite the limited computational abilities. TinyML optimizes the efficiency of robots, saving electric power, allowing robots to be deployed independent of connections to the cloud, and expanding the domain of autonomous robots to include resource-limited environments.

Hardware

We are using Nvidia JetBot for model deployment. It is powered by Jetson Nano Module that has:

- 128-core NVIDIA Maxwell™ GPU
- Quad-core ARM® A57 CPU
- 4 GB 64-bit LPDDR4 Ram
- 136° field of view wide camera.
- Ubuntu 18.04
- 5-Watt power supply
- WiFi and Bluetooth 4.0 adapter

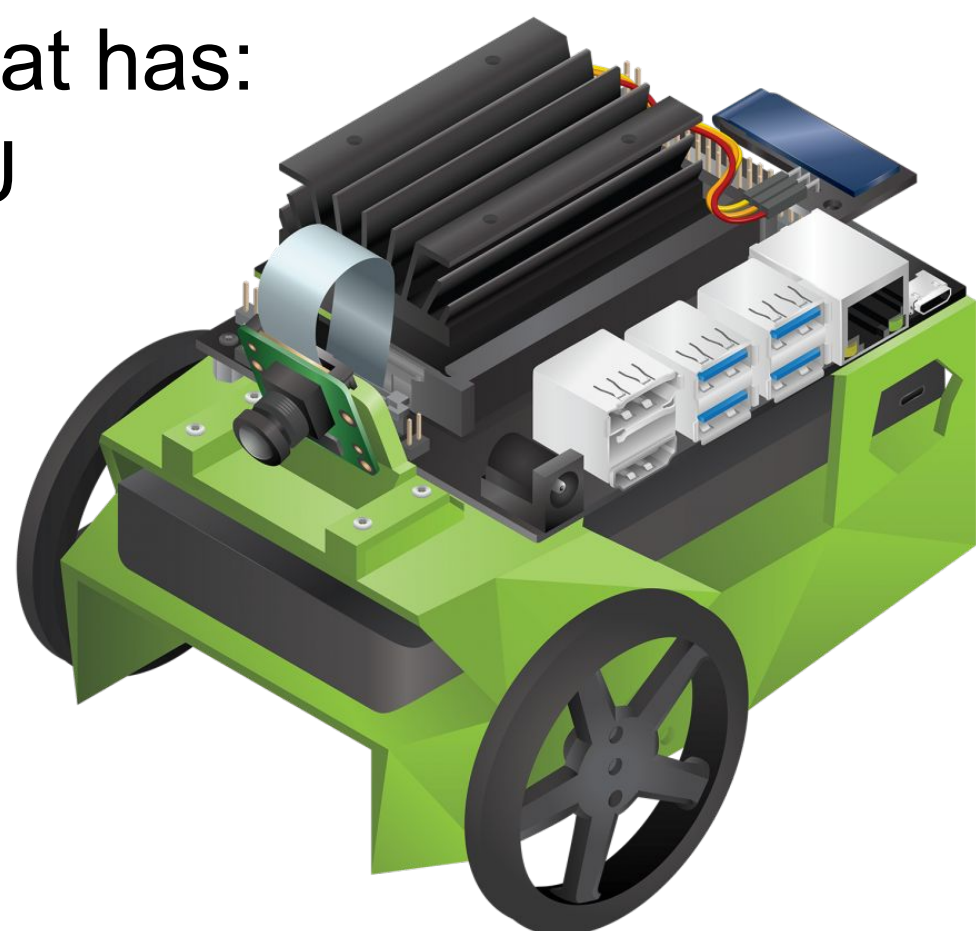


Figure 1. Nvidia JetBot 4 GB

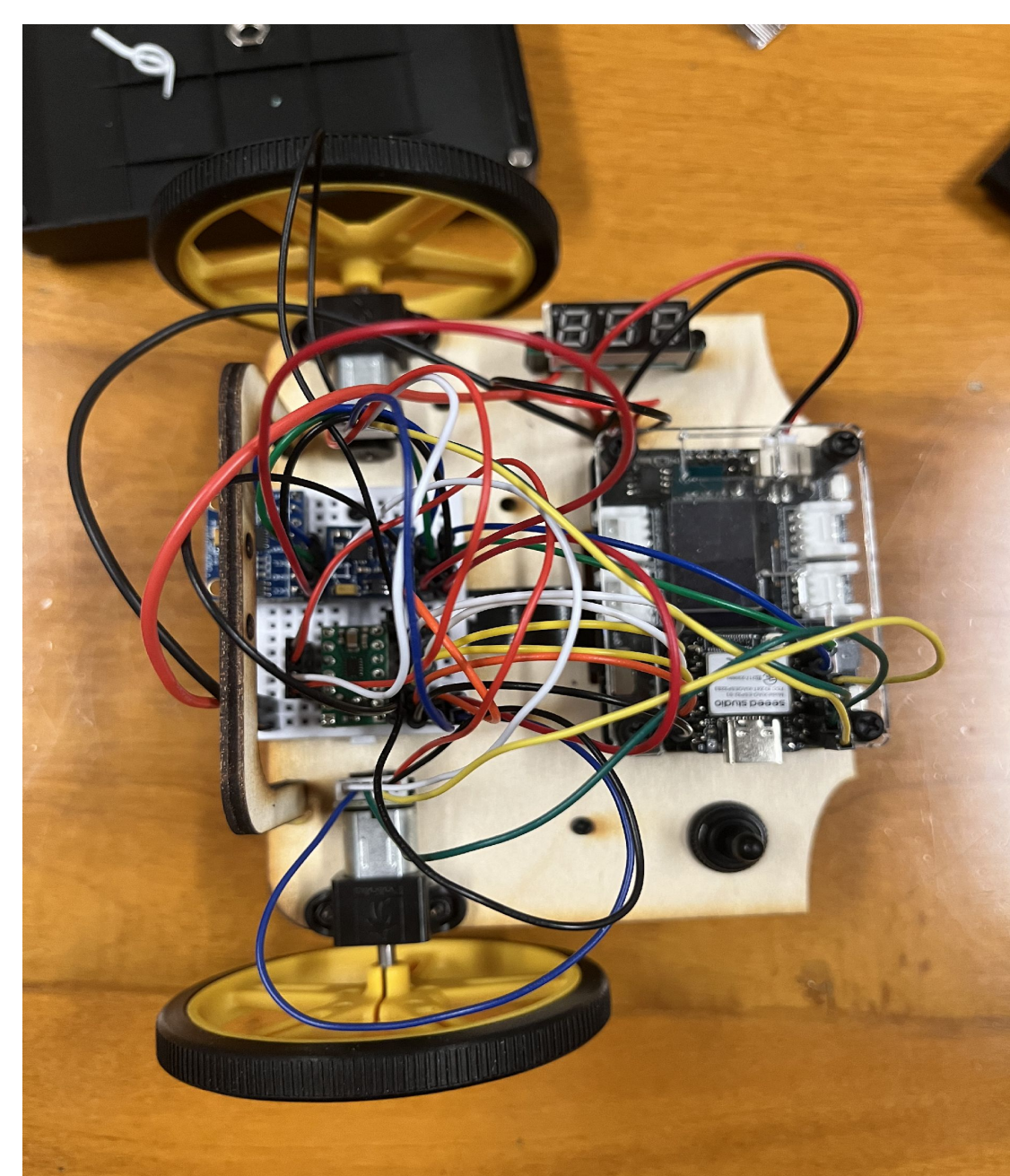


Figure 2. Picture of Robot

We are also using a custom robot that mainly consists of a laser cut wooden chassis, Pololu Wheels, and a 3D printed battery holder. A seeds Studios EPS32 microcontroller is used in combination with the Pololu Dual Motor Driver Carrier and a Pololu Motor. All coding with the microcontroller is done through Arduino, and the websites are built with a mixture of html and JavaScript.

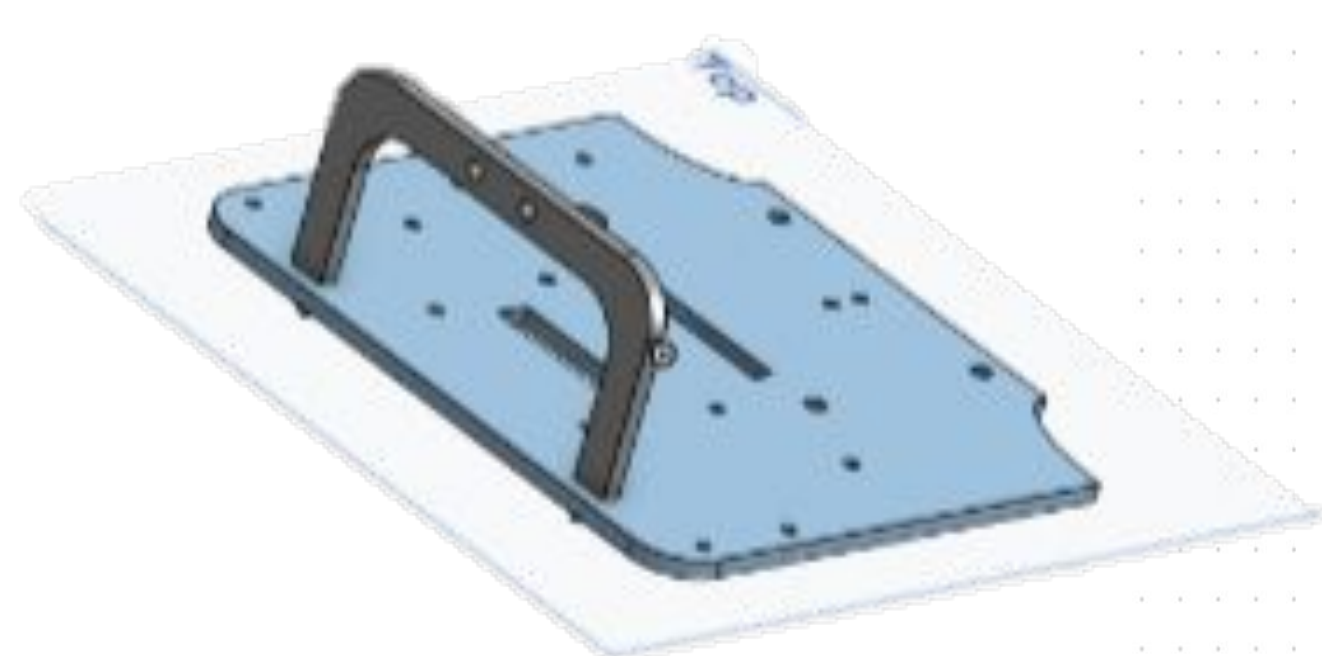


Figure 3. Laser Cut Chassis and LiDAR Arch

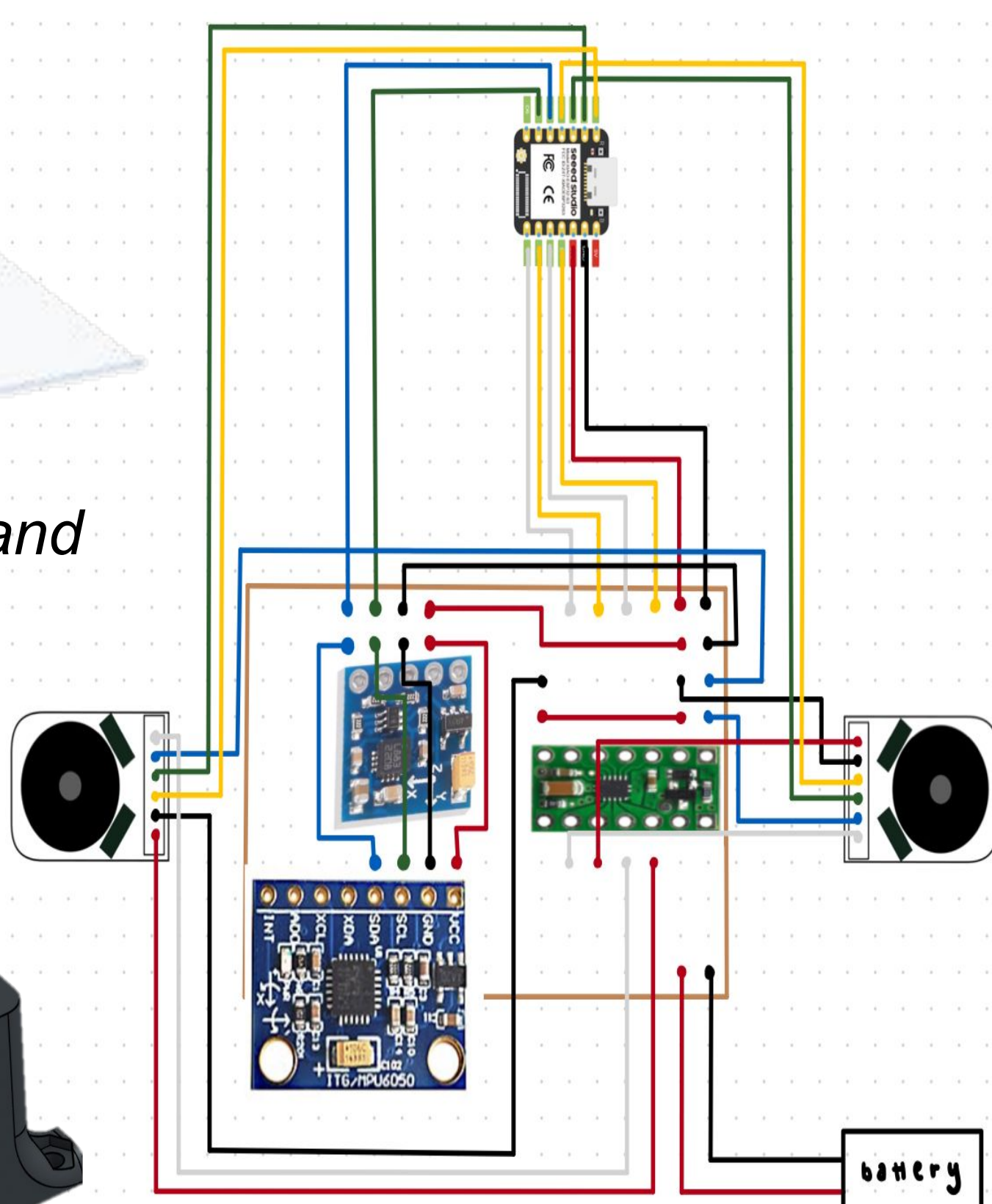


Figure 4. Robot Wiring Diagram

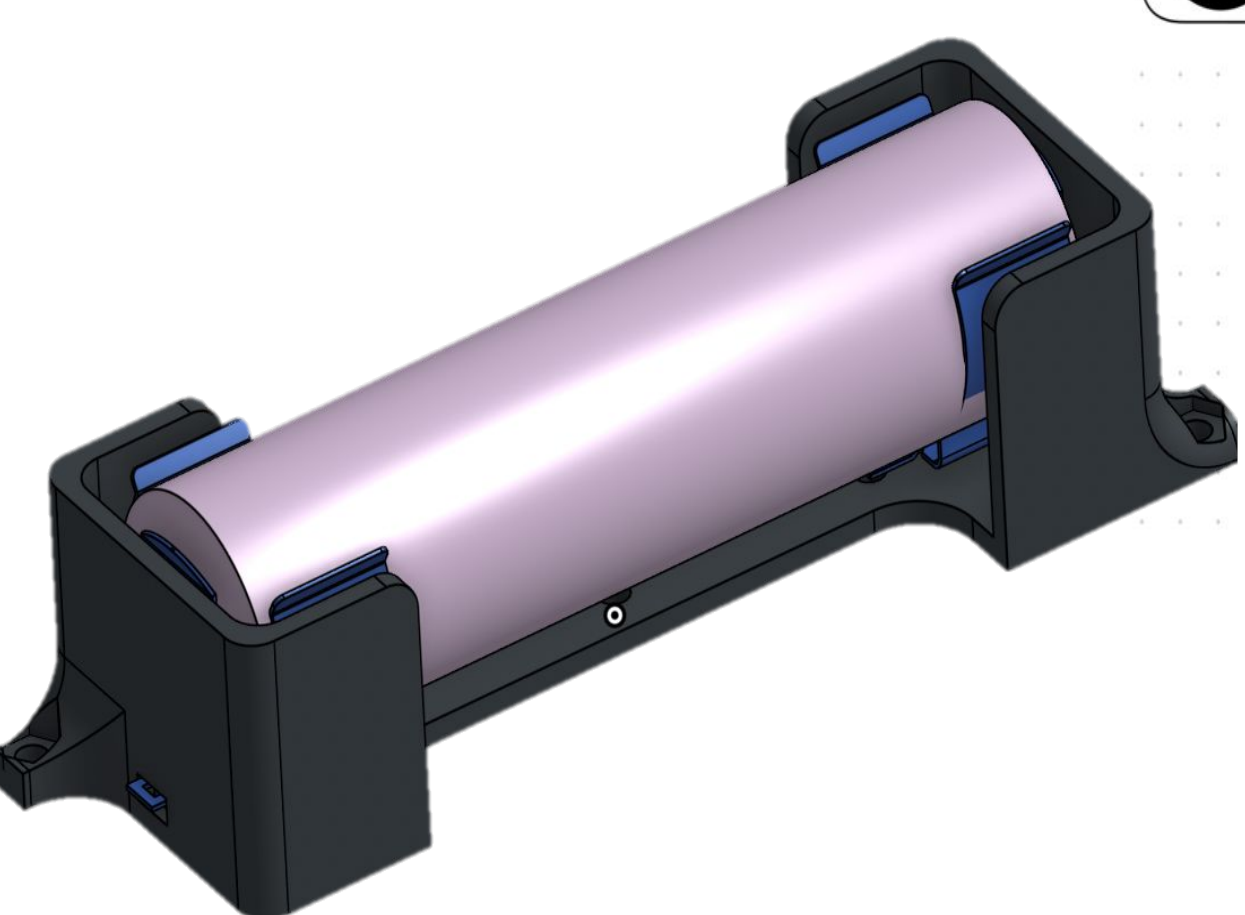


Figure 5. 3D Printed Battery Holder

Objectives

The objectives of the custom robot is to create a simple robot that is easy to assemble and serves as a good instructional model for teaching College students about the basics of robotics. This robot will play an important role in the upcoming Mobile Robotics course at Pomona College, taught by Professor Clark. The course is designed for computer science students who do not have a strong background in Engineering or Electronics.

Components of the Custom Robot

Bluetooth

- Created a webpage to control a robot via Bluetooth Low Energy (BLE).
- Users can control the speed of both wheels separately for movement and direction.
- Upon disconnection, both wheels stop automatically.
- Used Arduino BLE library to create and advertise a BLE server
- Initial problems:
 - BLE characteristics sent as strings led to dynamic memory allocation problems
 - Tried Arduino's BLE Typed Characteristics library but it was incompatible with other necessary BLE libraries.
- Switched to NimBLE library by @h2zero on GitHub, which supports integer and float BLE characteristics, resolving memory allocation issues.
- Regular Bluetooth scans cannot locate the device, but this is not a problem as connection is only through a specialized webpage.

Over-The-Air Updates

- OTA updates enable wireless edits and uploads to our Arduino script.
- Based on gb88's BLEOTA library on GitHub: <https://github.com/gb88/BLEOTA>.
- BLEOTA library is for BLE, not NimBLE, resulting in slower upload speeds with NimBLE.
 - NimBLE files are smaller (around 513 kb) compared to BLE (around 895 kb).
 - Upload speed with NimBLE is roughly 3 kb/s versus 10.5 kb/s with BLE.
- Despite slower updates, the small file sizes make this issue insignificant for our use.

Proportional Control

- The robot is encoded with a simple proportional control system, which allows the speed of the wheels to gradually increase.
- Prevents the motor current from getting too high, avoiding potential damage.
- Enhances the longevity of the robot by reducing wear and tear on the motors.

OLED Display and Speaker

- The expansion board for the ESP32 micro-controller includes OLED display screen and speakers
- OLED display have been coded to show connection status.
- Speakers chirp on connection or disconnection.
- Speaker pin conflicts with wheel encoder pins, which led us to temporarily abandon it for now

Custom Robot Future Direction

Moving forward, we will improve and debug the current robot, fixing issues such as the speaker pins and the efficiency of OTA upload. We will also create more guides and diagrams for the students to follow while building and coding their own robot.

Research Question

This project is part of a greater ARCS lab project that aims to build a robot that can autonomously navigate through Oldenburg residence hall at Pomona College. A lab team is developing DNN models that output a motion direction, "forward", "left", or "right", based on simulation-generated images of the hall.

The project's objective is to deploy the models on the Nvidia Jetbot, enabling it to use its camera for real-time image input, run inferences, and move based on the model's output. The research question is: How can we improve the model's accuracy with real image inputs?



Figure 6. Oldenburg Hallway

Methodology

Remote Procedure Calls:

To overcome the incompatibility of newer PyTorch and Fast.ai versions that were used in model development with Python 3.6, which is the latest version supporting the Nvidia Jetbot, we built two separate environments: one for model execution and one for robot control, communicating via Remote Procedure Calls (RPCs). The Jetbot captures an image, saves it locally, and calls the model inference function with the image filename. The model loads the image, runs the inference, outputs a motion direction, and then calls the robot motion function with that direction. .

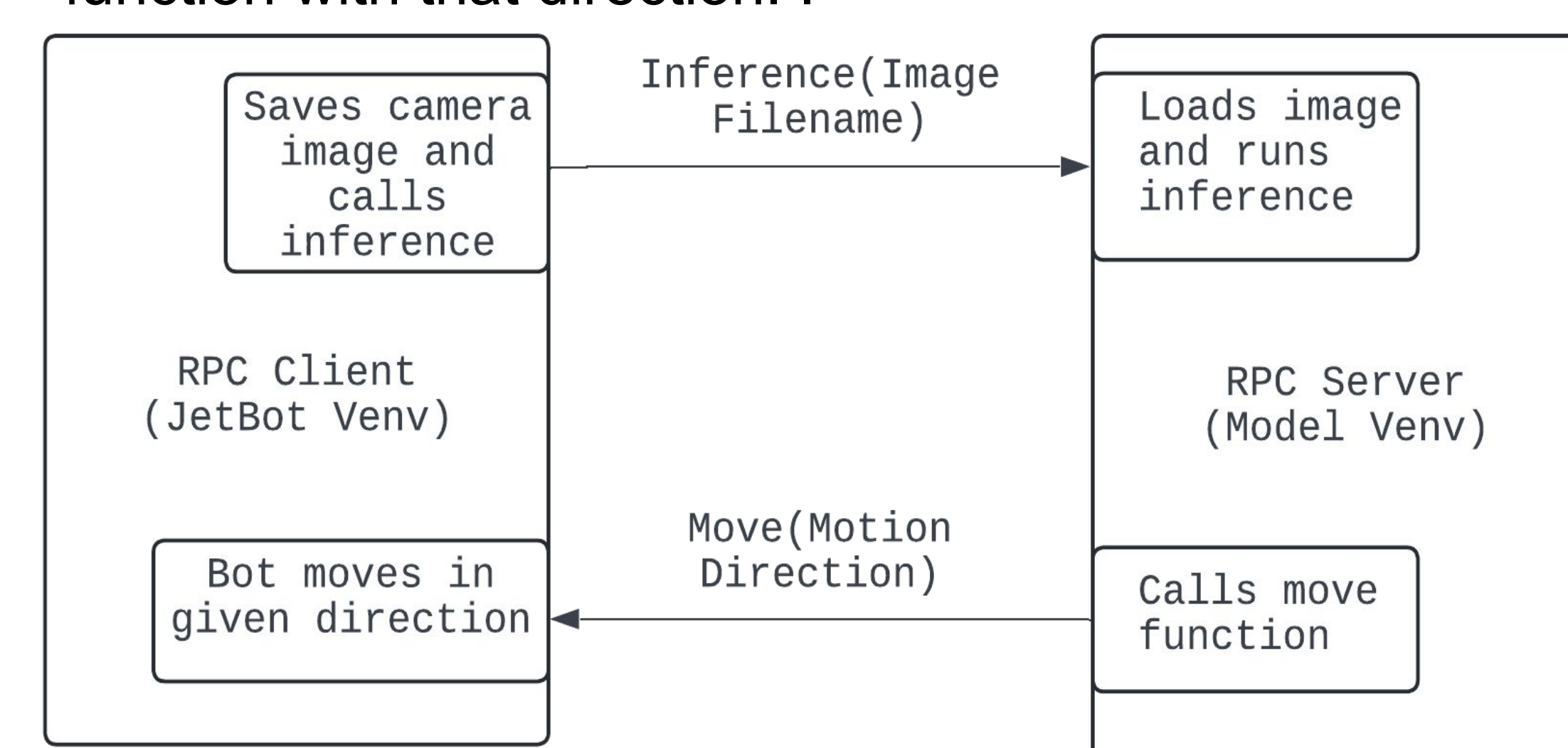


Figure 7. Jetbot/Model Remote Procedure Calls

Camera Calibration:

To overcome the camera's distortion, which causes straight lines to appear curved and parts of the image appear closer than reality, we calibrated the camera, using checkerboard images to calculate the distortion coefficients and correct them. The following image is taken after and before camera calibration.

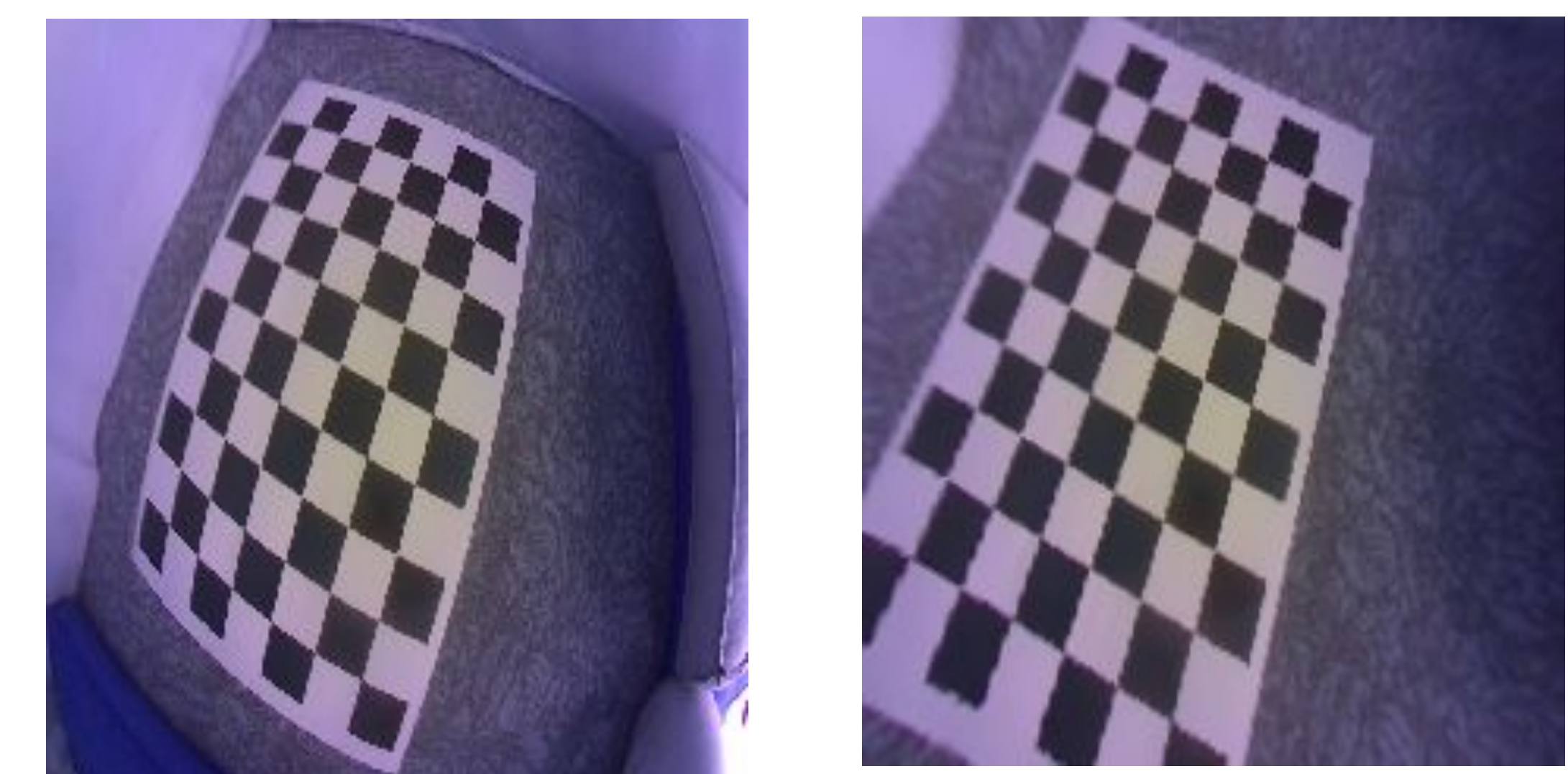


Figure 8. Checkerboard before and after calibration

Future Experiments

We will run experiments with the JetBot in Oldenburg to gather real data for training and validation and improve the model accuracy. Also, we will test the model in various scenarios with unexpected obstacles, such as humans showing up, to improve the model reliability.